

# Physics-inspired coarse-scale perturbation

Phase 3 ablation: target the parameters at wavelengths  $\lambda \gtrsim 3000$  km (upper synoptic and above). A different mechanism per model, same physical objective.

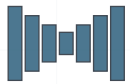
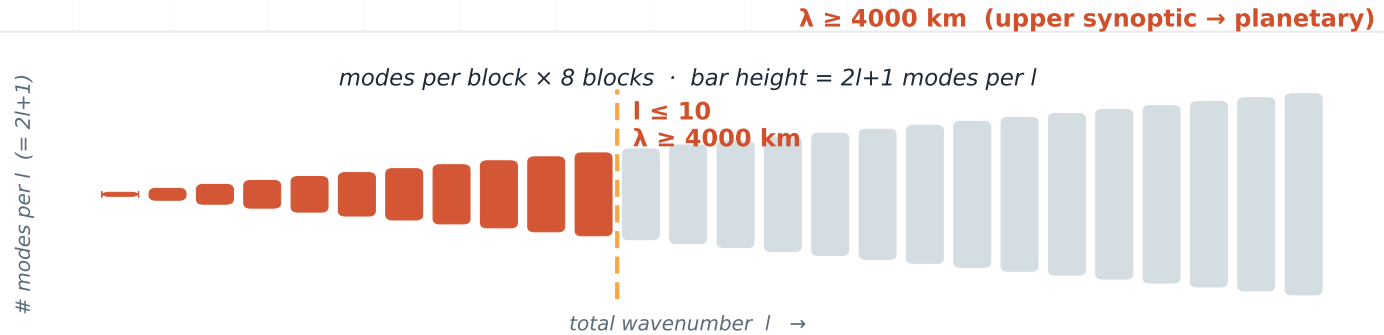


## SFNO

Spherical Fourier Neural Operator

**Mechanism:**  
spectral weight sub-slice perturbation

**Target slice:**  
 $l \leq 10$  modes of every block's filter weight

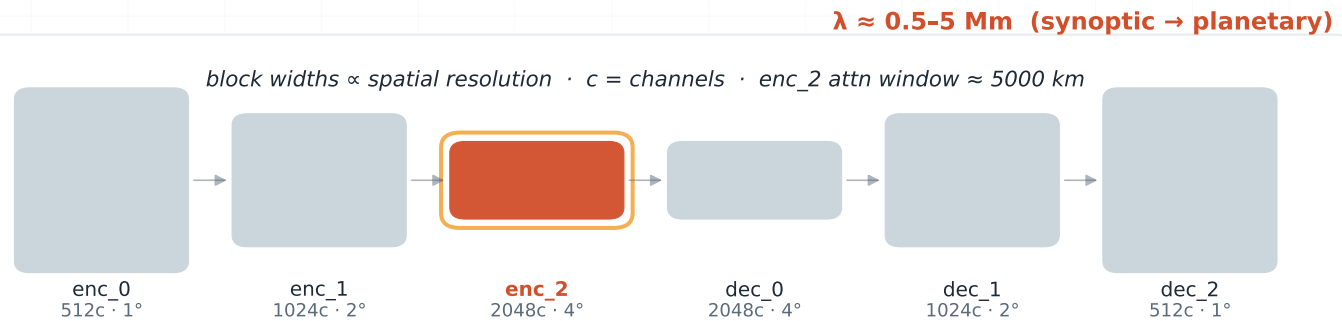


## Aurora

Swin-Transformer 3D U-Net

**Mechanism:**  
U-net-bottom weight perturbation

**Target slice:**  
encoder\_layers.2 only

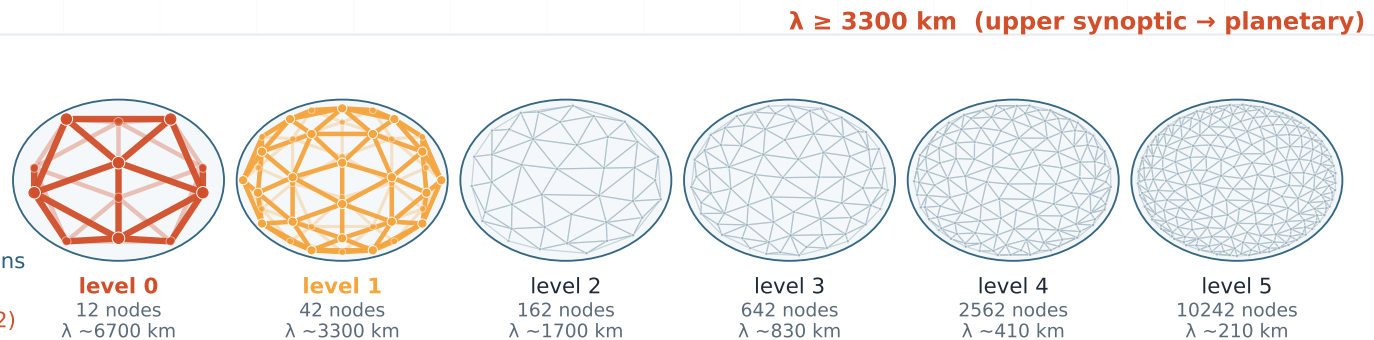


## GraphCast

Multi-Mesh Graph Neural Network

**Mechanism:**  
runtime hook on coarse mesh-node activations

**Target slice:**  
mesh nodes at refinement levels 0-1 (first 42)



● targeted (perturbed)    ● intermediate scale    ● untouched (small-scale)